

How many years have passed? : Comprehensive Wear Analysis and Simulation Models for Stairs

Summary

Due to the nature of walking patterns, wear on stairs is uneven. Factors such as stair shape, environmental conditions, and the interaction between natural and human-made elements also influence this wear. We considered key factors to establish a wear model based on a realistic physical framework. Additionally, relevant models were developed to identify and determine the specific parameters needed for the overall wear model. By inputting assumed values for unknowns, we simulated wear patterns and compared them for accuracy to address the related issues.

Model 1: Holistic Wear Model: Based on a real physical model, considering Archard wear, fatigue wear, thermal expansion wear, chemical wear and their nonlinear coupling terms, a forward simulation wear model can be built.

Model 2: Walking Pattern Determination Model: 3D stair data can be obtained via laser point cloud and orthophoto technologies, and Finite Element Analysis (FEA) can help discretize the data we get from scanning. And then, by utilizing the Random Forest Model, we can classify the travel mode on the steps.

Model 3: Parameter Optimization Model: By using residual analysis, we calculated and analyze the Sum of Squared Residuals(SSR). And then we are able to fit unknown parameters and optimize known parameters. This will provide more diverse and reliable data to get more precise simulation results through the Holistic Wear Model.

Model 4: Year Estimation and Renovation Determination Model: The ideal stair construction year can be estimated by using the Central Limit Theorem to analyze the difference between each simulated wear condition with the scanned real wear condition. And Goodness-of-Fit tests are applied to analyze whether the stairs have undergone partial or complete renovations or both.

Finally, we tested the sensitivity and reliability of the models, analyzed their strengths and weaknesses. Also, we proposed that potential future research can be extended to scenarios like steps that have both flat surface and uneven patterned surface on different spots. Moreover, floors with even or uneven patterns can also be our potential research objects. By expanding our model on them and conducting wear analysis on them, we can explore relevant social and historical contexts behind them.

Keywords: Wear; Travel mode; Archard wear model; Fatigue wear model; Laser point cloud and orthophoto technology; FEA analysis; Random Forest model

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1 Introduction

1.1 Problem Background

In recent years, with the continuous development of computer technology, big data and artificial intelligence technology have become more and more popular and have been widely used in various fields. Among them, archaeology, which seems to be the most alienated from computer technology, is no exception. At present, various computer technologies have been widely used in the field of archaeology. For example, laser point cloud and orthophoto technology can be of great help to field data survey. At the same time, in recent years, due to the continuous increase in computing power, archaeologists have begun to use computers to build models to help study the wear on the surface of cultural relics, so as to restore cultural relics.

This has also inspired archaeologists to think about whether they can study the age of the cultural relics themselves and the historical and social changes they have witnessed by exploring the wear on the surface of the steps of these cultural relics.



Figure 1: 3D real scene of cultural relics in China

1.2 Restatement of the Problem

Considering the background information and restricted conditions identified in the problem statement, we need to solve the following problems:

- Determine what measurements we should make and ensure them to be economical and accessible.
- Study the travel patterns of people using the stairs. This mainly refers to the certain direction and distribution of their travel pattern. For example, do they primarily ascend, descend, or move bilaterally? Do they travel in single file or side by side?
- Determine the number of people using the stairs.
- Analyze whether your simulated wear from your model is consistent with the measured wear when you have enough parameters.
- Predict the possible age of the stairs and calculate reliability of this prediction.
- Determine whether the stairs experienced repairs or renovations. If yes, classify it into partial or total repairs/renovations.

- Determine the possible source of material by analyzing if they have similar wear results.

1.3 Our Work

According to the requirements, our work is as follows:

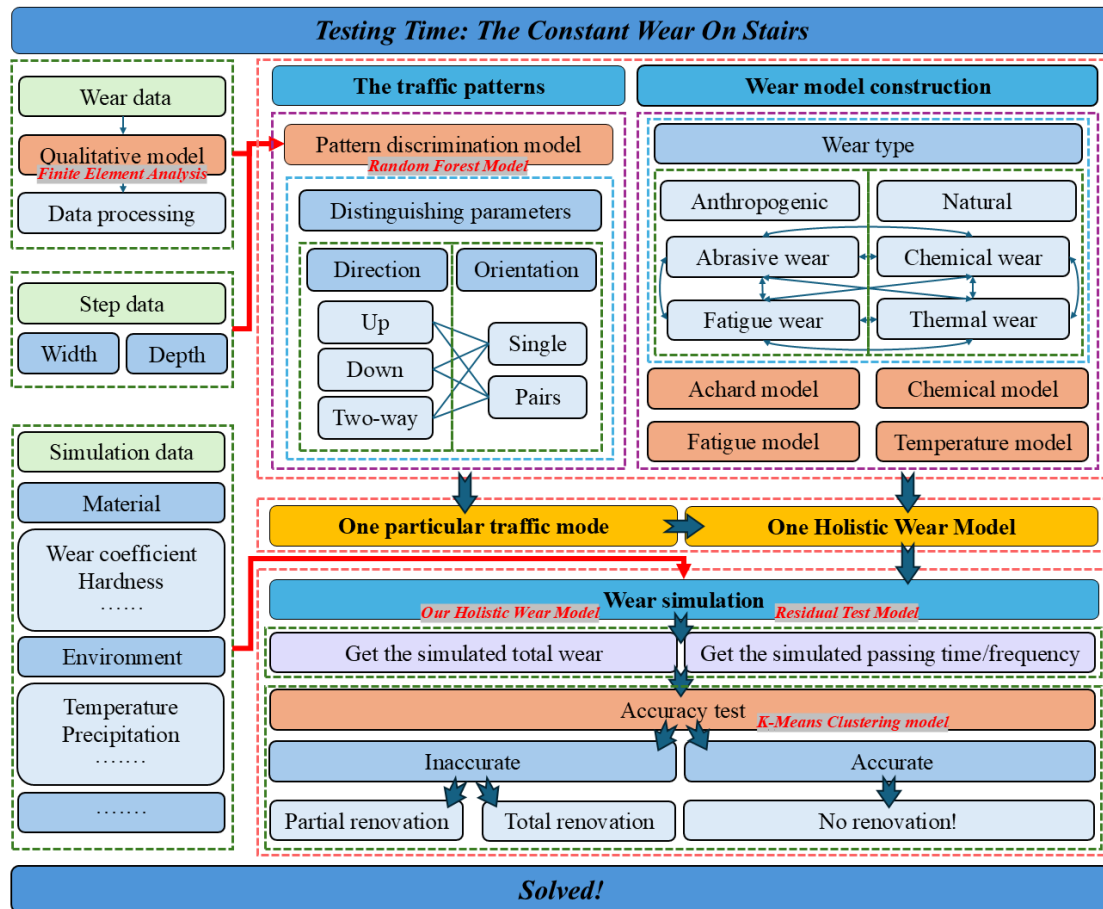


Figure 2: Our Work

2 Assumptions and Justifications

Considering those practical problems always contain many complex factors, first of all, we need to make reasonable assumptions to simplify the model, and each hypothesis is closely followed by its corresponding explanation:

- **Assumption 1:** All the data we use is accurate and valid.

■ **Explanations:** In real-life research, we are able to collect data from field survey, laboratory experiment and online official databases, including the hardness of the rock or wood, the wear coefficient of the rock or wood, the temperature and precipitation distribution of the environment. While it might seem idealistic, this assumption is reasonable within the context of scientific modeling.

- **Assumption 2:** The steps we studied all have flat surfaces at its initial state.

■ **Explanations:** Newly constructed stairs typically do have relatively flat surfaces, thus convenient for people to walk on. Moreover, this assumption makes it easier for us to measure and determine the total wear, and thus conduct the required research.

● **Assumption 3:** The average human weight is 62kg and the average foot length is about 25cm and the average shoulder width is 40cm.

■ **Explanations:** The average human weight and foot length are significant to determine the normal direction load and stress, further influencing the abrasive and fatigue wear per time. The use of average values is a reasonable approach in mathematical modeling, as it allows us to represent typical conditions without getting bogged down in individual variations. This simplification is crucial for creating a model that can be broadly applied while still capturing the essential aspects of how human interaction contributes to stair wear. To simplify the model, we ignore the changes in these factors and choose some typical values.

● **Assumption 4:** The soles of human shoes and feet are flexible and can deform when walking on stairs.

■ **Explanations:** Throughout history, most regions of the world have predominantly used flexible materials for shoe soles, with the exception of specific areas and periods (such as the use of wooden clogs in certain eras in China). The wear patterns on stairs differ significantly between flexible and rigid shoe soles.

● **Assumption 5:** The material composition of pedestrians' shoe soles remains constant for the stairs under study.

■ **Explanations:** This assumption is grounded in historical continuity and technological constraints. Throughout much of human history, shoe sole materials have remained relatively consistent within specific geographical and cultural contexts over extended periods. This assumption allows us to simplify models of stair wear, focusing on the cumulative effect of a consistent type of shoe sole over time, rather than accounting for rapid or frequent changes in material composition.

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Definition	Unit
L	Front edge of step	-
D	Whole region of the steps	-
A, B	Two feet of one walker	-
O	Origin point (The center of the steps)	-
x, y, z	Coordinate parameters	dm
ΔV	Volume of wear	dm ³
$W(x, y)$	Work of wear	J
$w(x, y)$	Work of wear per unit area	J/dm ²
Δz	Total wear height within an area	dm
dz	Wear height per unit area	dm
X	$X = \frac{\iint_D x * w(x, y) dx dy}{\iint_D w(x, y) dx dy}$	dm

Y	$Y = \frac{\iint_D y * w(x, y) dx dy}{\iint_D w(x, y) dx dy}$	dm
$\overline{X}$	$\overline{X} = \frac{1}{n} \sum_{i=1}^n X_i$	dm
$\overline{Y}$	$\overline{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$	dm
H	Hardness of the step material	Pa
K	Coefficient of Archard wear	-
$p(x)$	(Marginal) probability density function	-
$p(x, y)$	Joint probability density function	-
$N(\mu, \sigma^2)$	Normal distribution	-
$\aleph(\mu \pm \nu, \sigma^2)$	$p_{\aleph}(x) = \frac{1}{2} (p_{N_1}(x) + p_{N_2}(x))$ $N_1 \sim N(\mu + \nu, \sigma^2), N_2 \sim N(\mu - \nu, \sigma^2)$	-
$f(x, y)$	Stair function scanned in 3D	-
$g(x, y)$	Simulated stair function	-
SSE	Sum of Squares for Error	-

4 Model establishment

4.1 The Holistic Wear Model

4.1.1 The proposal of the holistic wear model

In real life, wear is a complex process resulting from many different factors, both anthropogenic and natural. So, we say, total wear is a combination of individual wear types and their interactions. Here, we consider four main wear types: abrasive wear, fatigue wear, thermal weathering wear and chemical wear.

And there're complex interactions between these wear types. For example, fatigue wear and thermal weathering wear cause cracks on the stair surface, reducing material strength, thereby making the stairs more susceptible to abrasive wear; they also accelerate the rate of chemical dissolution.

So, now we will have an insight into these different types of wear.

4.1.2 Two-body abrasive wear

In order to build the model, first, we need to know how wear happens; next, for a single step, we can find a physics principle; then, when large number of steps accumulate, statistics can be applied to convert the problems; finally, two parameters and distributions can be subtracted.

I A Single Step: Wear Pattern

“Feet” hereafter represents the objects (feet or shoes) that wear the stairs. Assume feet are soft enough to deform when walking on the stairs.

Two-body abrasive wear occurs when feet move relative to the stairs. The wear is related to the depth of the stairs and the walking mode.

→ For stair depth ■ For walking modes

→ For a step of normal depth, in a stationary state, the sole of the foot can be fully placed on the step. The wear patterns differ when ascending and descending:

■ Ascending

- Foot A steps onto the step first.
- The center of gravity shifts to leg A.
- The heel of foot B lifts, and its front sole experiences slight displacement with the step.
- The step region $\{(x, y) | 0 < y < b/2, (x, y) \in D\}$ undergoes two-body abrasive wear.
- The heel of foot A may contact and rub against the step edge L , causing additional wear.

■ Descending

- Foot A steps down first.
- The front sole of foot B rubs against the step region $\{(x, y) | -b/2 < y < 0, (x, y) \in D\}$.
- The front sole of foot B almost always rubs against the step edge L .

Therefore, the main abrasive wear areas are:

■ Ascending

$$D_1 = \{(x, y) | 0 < y < b/2, (x, y) \in D\} \cup L$$

■ Descending

$$D_2 = \{(x, y) | -b/2 < y < 0, (x, y) \in D\} \cup L$$

→ For a step of shallow depth, half of the sole is suspended.

■ Ascending

Because the front sole can still contact the step, the wear type is the same as that of normal depth.

■ Descending:

- Only heel of foot B can contact the step surface, so friction from the front sole no longer occurs.
- Foot B can still rub against the step edge L .

II A Single Step: Two-body Abrasive Wear

Two-body abrasive wear can be described by *Archard's law*. (Moore, 1974) When wear happens within the area $A_{wear} = [x, x + \Delta x] \times [y, y + \Delta y]$, the volume of wear can be described by

$$\Delta V = \Delta x \Delta y \Delta z = \frac{K}{H} W(x, y). \quad (1)$$

- $W(x, y) = Fl$: frictional work

- F : pressure at the point of friction

- l : sliding distance

- H : material hardness

- K : wear coefficient

According to the mechanism of two-body abrasive wear, abrasion occurs only at the point of contact between one foot and the step, while the other foot supports the body. During the abrasion process, the person experiences acceleration. Consequently, the pressure F is not equal to the person's body weight G .

We make the following assumptions and parameter selections:

- $F = 0.3G$ (This assumption requires further experimental validation)

- Average body weight $G = 62$ kg (based on international average)

- Sliding distance $l = 0.1$ mm

It's important to note that the material hardness H depends solely on the properties of the object being abraded. The wear coefficient K is determined by the pairing of the abrasive and abraded materials, as well as the smoothness of the

contact surface. Both H and K need to be determined through experimental measurements for accurate model predictions.

When $A_{wear} \rightarrow 0$, wear happens at a certain point (x, y) . Since wear only occurs in the z direction, the wear height per unit area is

$$dz = \frac{K}{H} w(x, y), \quad (2)$$

$$\Delta z_{2Body} = \frac{K}{H} \frac{W(x, y)}{\Delta x \Delta y} = \iint_A dz(x, y) dx dy, \quad (3)$$

where $w(x, y) = \partial^2 W / \partial x \partial y$ is the work per unit area.

III Statistics: Problem Conversion

Since the notable stair wear is the result of a large number of people using the stairs, the process can be described using statistical language. The total wear caused by numerous steps on an area of the stairs can be equivalently modeled as follows: first, we concentrate the total wear from a single step at one point, then randomly select a large number of points within this area, and apply the total wear of a single step to each of these points. This approach allows us to capture the cumulative effect of many individual steps.

□

The mathematical expressions are:

Define random variables X and Y representing the coordinates of a single step as

$$X = \frac{\iint_D x * w(x, y) dx dy}{\iint_D w(x, y) dx dy}, \quad (4)$$

$$Y = \frac{\iint_D y * w(x, y) dx dy}{\iint_D w(x, y) dx dy}, \quad (5)$$

i.e., (X, Y) is the center of work, which is an analogy to the center of mass. Actually, (X, Y) represents the point with the most significant wear in a single wearing process. Different individuals have different walking habits. (X, Y) follows a certain bivariate distribution F_{XY} , with the joint probability density function $p(x, y)$.

Different people's stepping on the stairs constitutes independent repeated experiments, forming n random points in D . $\{(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)\}$ represents the locations of the random points, forming samples. $(X_i, Y_i) (i = 1, 2, 3, \dots)$ can be treated as independently, identically distributed random variables, each distributed as (X, Y) , with the same finite mean μ and variance σ^2 . This assumption is reasonable because individual steps on the stairs can be considered independent events, and the overall pattern of wear is likely to follow a consistent distribution during a period of time, which will be further discussed below.

Define the total wear function $dz_{2Body}(x, y)$ at any point (x, y) in D as (to concentrate the total wear from a single step at one point)

$$dz_{2Body}(x, y) = \Delta z * \sum_{i=1}^n \delta(x - X_i, y - Y_i), \quad (6)$$

where $\delta(x, y)$ is the Dirac delta function, which is used here to represent the concentrated wear at a single point, and (X_i, Y_i) are the coordinates of the i -th random point (to randomly select a large number of points within this area).

The expected wear at any point (x, y) in D is

$$E[dz_{2Body}(x, y)] = \Delta z * n * p(x, y). \quad (7)$$

Thus, the total expected wear over the entire area D is

$$E\left[\iint_D dz_{2Body}(x, y) dx dy\right] = \Delta z * n. \quad (8)$$

In practice, the integration can be calculated by the summation by large amount of data, corresponding to numerous people stepping on the stairs. This approximation bridges the gap between our theoretical model and real-world observations:

$$E\left[\lim_{n \rightarrow \infty} \sum_{i=1}^n dz_{2Body}(X_i, Y_i) \Delta X_i \Delta Y_i\right] = \Delta z * n. \quad (9)$$

The left part is to apply the total wear of a single step to each of these points, while the right part is the origin total wear caused by numerous steps.

□

IV Statistics: Use Theorem, Find Parameters

The factors affecting X and Y can be categorized into individual, collective, and environmental factors. These factors are integrated into our model through their influence on the probability distributions of X and Y .

- the walking habits individually
- the walking modes collectively
- the stair shape, which includes:
 - stair width and depth
 - surface condition

The surface condition exerts influence through the following way: Wear, especially wear at the edges, will cause a change in the location where wear occurs. In other words, the former distributions $F_{XY}(t_i)$ will affect the later distributions $F_{XY}(t_i + \Delta t_i)$, where Δt_i is relatively large time interval, such that the morphology of the steps undergoes significant changes from t_i to $t_i + \Delta t_i$. Thus, $F_{XY}(t)$ should be a piecewise function of time t , with Δt_i as the length of the i -th interval of the piecewise function.

During the time period $t \in [t_i, t_i + \Delta t_i]$, assume X and Y are independent. As discussed above, X_i and Y_i can be treated as independently, identically distributed random variables. When a large-scale population steps on the stairs, i.e., a large number of independent repeated experiments are conducted, the *Central Limit Theorem for Independent and Identically Distributed Random Variables* can be applied. That is: regardless of the unknown distributions F_X and F_Y , as the sample numbers approaches infinitely, the sample mean $\bar{X}$ and $\bar{Y}$ converge to normal distributions.

□

The mathematical expressions are:

Let $X_1, X_2, \dots, X_n$ be independently, identically distributed random variables with $E(X_i) = \mu$ and $Var(X_i) = \sigma^2 < \infty$.

Define

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad (10)$$

Then

$$\bar{X} \xrightarrow{d} N\left(\mu, \frac{\sigma^2}{n}\right) \quad (11)$$

Under the different constraints, like the collective walking modes or stair shape, F_X and F_Y have different finite mean μ . Thus, the sample mean $\bar{X}$ and $\bar{Y}$ follow normal distributions, with corresponding parameters, i.e., $\bar{X} \sim N(\mu_1, \sigma_1^2)$ and $\bar{Y} \sim N(\mu_2, \sigma_2^2)$.

□

V Statistics: Use Constraints, Find Distributions

Below will different constraints be discussed. For simplification, the discussion is started from the initial state, where the stair surface is flat, and the surface condition is not taken into consideration, temporarily.

→ For stair shape (width and depth only) ■ and ● For walking modes

The depth of the steps will constrain F_Y (the wear within D).

→ Constraints of normal depth steps

■ Ascending

Given the discussion of how people walk downstairs or upstairs, when ascending, footfall concentrates in the middle of $\{(x, y) | 0 < y < b/2, (x, y) \in D\}$. If the stair is an infinite plane, the distribution of Y , i.e., F_Y , should be symmetric around $y = b/4$. Thus, $\mu_2 = b/4$, $\bar{Y} \sim N(b/4, \sigma_2^2)$. Sample $Y_i (i = 1, 2, 3, \dots) \in [b/2, -b/2]$, thus $\bar{Y} \in [b/2, -b/2]$. We choose σ_2^2 such that $P(\bar{Y} \in [b/2, -b/2]) = 95\%$ ($\sigma_{1,2}^2$ hereafter represents values processed this way).

■ Descending

Footfall concentrates in the middle of $\{(x, y) | -b/2 < y < 0, (x, y) \in D\}$. $\mu_2 = -b/4$, $\bar{Y} \sim N(-b/4, \sigma_2^2)$.

→ Constraints of shallow depth steps

■ Ascending

$$\mu_2 = b/4, \bar{Y} \sim N(b/4, \sigma_2^2).$$

■ Descending

Because the friction of the front sole does not exist, no wear occurs.

The width of the steps will constrain F_X .

→ Constraints of single-shoulder width steps

Footfall concentrates in the central area of regions D and L . As both feet cause wear, the means of F_X for the two feet are $\mu_{1right} = \delta/2$ (right foot) and $\mu_{1left} = -\delta/2$ (left foot), respectively. $\bar{X}_{right} \sim N(\delta/2, \sigma_1^2)$, $\bar{X}_{left} \sim N(-\delta/2, \sigma_1^2)$. The distribution of $\bar{X}$ is obtained by superposition and normalization of the two feet distributions, thus $\bar{X} \sim \aleph(\pm \delta/2, \sigma_1^2)$.

→ Constraints of double-shoulder width steps

● High short-term foot traffic

Assume people conventionally walk on their right side.

■ Ascending

Footfall concentrates in the middle of $\{(x, y) | 0 < x \leq a/2, (x, y) \in D\}$. $\mu_1 = \pm\delta/2 + a/4$, $\bar{X} \sim \aleph(\pm\delta/2 + a/4, \sigma_1^2)$.

■ Descending

Footfall concentrates in the middle of $\{(x, y) | -a/2 < x \leq 0, (x, y) \in D\}$. $\mu_1 = \pm\delta/2 - a/4$, $\bar{X} \sim \aleph(\pm\delta/2 - a/4, \sigma_1^2)$.

● Consistently low foot traffic:

Footfall is assumed to concentrate in the central area of regions D and L . $\mu_1 = \pm\delta/2$, $\bar{X} \sim \aleph(\pm\delta/2, \sigma_1^2)$

□

Note: Define the $\aleph(\mu \pm \nu, \sigma^2)$ distribution as

$$p_{\aleph}(x) = \frac{1}{2} (p_{N_1}(x) + p_{N_2}(x)), \quad (12)$$

where $N_1 \sim N(\mu + \nu, \sigma^2)$, $N_2 \sim N(\mu - \nu, \sigma^2)$ and $p_{\aleph, N_1, N_2}$ is the probability density function of $\aleph, N_1, N_2$ respectively.

□

Below will the surface condition be discussed (the wear on L).

Initially, $Y = b/2$. The wear process begins with a rapid rate. As wear progresses, the point $b/2$ erodes, exposing points within the interval $[b/2 - \varepsilon(t), b/2]$ to wear. Here, $\varepsilon(t)$ is a monotonically increasing function of time t .

The wear process transitions to a probabilistic model where Y fits a certain distribution with the property:

$$P(b/2 - \varepsilon(t) \leq Y \leq b/2) = 1 \quad (13)$$

While it may seem intuitive to choose the midpoint of the interval as the mean μ_2 , this approach presents challenges:

- The changing interval size over time increases computational complexity.
- Wear is most significant in the neighborhood of $b/2$.

Considering that the majority of a normal distribution's probability mass lies within a small neighborhood of its mean, we propose

- Set $b/2$ as the mean of the normal distribution.
- Replace $\varepsilon(t)$ by a small constant $b/64$. Select the standard deviation σ^2 such that:

$$P(b/2 - b/64 \leq Y \leq b/2) = 0.95 \quad (14)$$

This approach balances computational efficiency with the physical reality of the wear process.

4.1.3 Three-body abrasive wear

In arid environments such as deserts, outdoor stair surfaces accumulate debris particles, primarily silicon dioxide. This leads to three-body abrasive wear (shoes-particles-stairs) alongside two-body wear (shoes-stairs).

It can likewise be described by Archard's law.

$$dz_{3Body} = \frac{K'}{H} W(x, y) * \sum_{i=1}^n \delta(x - X_i, y - Y_i) \quad (15)$$

- K' : wear coefficient between silicon dioxide and rock. The frequency $\approx$ proportion of stair surface covered by debris (assumed 0.5 for demonstration).

4.1.4 Fatigue Wear

The periodic loading on stairs leads to fatigue and aging of the material, resulting in microcracks and surface peeling.

$$dz_{fatigue} = K'' * \sigma^s * n_f \quad (16)$$

- K'' : wear coefficient.
- σ : stress on the stair surface.
- s : stress sensitivity index.
- n_f : frequency of fatigue wear = 2*frequency of walkers.

4.1.5 Thermal Weathering Wear

In regions with significant diurnal temperature variations, for rock stairs, due to the low thermal conductivity of rock, temperature differences between the interior and exterior of the rock occur when environmental temperature changes, generating tensile stress, causing fissures in the rock and surface peeling.

$$dz_{thermal} = K''' \frac{\alpha * \Delta T}{H * \sqrt{c}} \quad (17)$$

- α : volumetric expansion coefficient.
- ΔT : temperature difference.
- H : hardness: hardness can be a measure of rock's resistance to damage, thus negatively correlated.
- c : specific heat capacity: larger specific heat capacity results in slower rock heating, smaller internal-external temperature difference, thus negatively correlated.
- K''' : proportionality coefficient (estimated 0.4). The weathering rate of granite in desert environments is approximately 0.5mm/1000 years, thus it alone has a minimal impact.

The wear occurs on approximately 250 days/year, on which days temperature differences are significant.

4.1.6 Chemical Dissolution Wear

In areas with acid rain, limestone or marble stairs can be dissolved by acid rain.(Perino & Talavera, 2014)

$$dz_{chemical} = \frac{k_0 * Con^a * V_{mol}}{A} \quad (18)$$

- k_0 : Chemical reaction rate constant.
- Con : Solution concentration (assume solution is sulfur acid).
- a : Reaction order.
- V_0 : Molar volume of calcium carbonate ($36.9 \text{ cm}^3/\text{mol}$).

4.1.7 Combined Wear Model

Total wear is a combination of individual wear types and their interactions. For example, fatigue wear and thermal weathering wear cause cracks on the stair surface, reducing material strength, thereby making the stairs more susceptible to abrasive wear; they also accelerate the rate of chemical dissolution.

Therefore, the total wear amount can be expressed as a linear combination of individual wear types plus the square root of their product (representing the coupling term, with square root to maintain dimensional consistency):

$$\begin{aligned}
 dZ_{human} &= dz_{2Body} + dz_{3Body} + dz_{fatigue} + \beta_{11}\sqrt{dz_{2Body}dz_{3Body}} + \beta_{12}\sqrt{dz_{2Body}dz_{fatigue}} + \beta_{13}\sqrt{dz_{3Body}dz_{fatigue}} \\
 dZ_{nature} &= dz_{thermal} + dz_{chemical} + \beta_2\sqrt{dz_{thermal}dz_{chemical}} \\
 dZ &= dZ_{human} + dZ_{nature} + \beta_3\sqrt{dZ_{human}dZ_{nature}}
 \end{aligned}
 \tag{19}$$

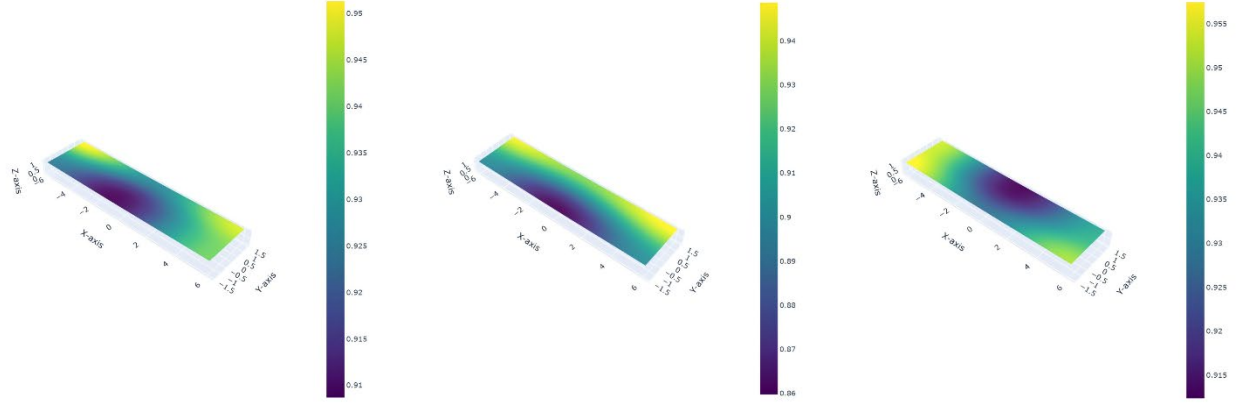


Figure 3: Simulated wear results (w=12.0dm, d=3.0dm, h=1.0dm, fn=20000/yr, year=200yr, material: granite, sole_material: rubber)

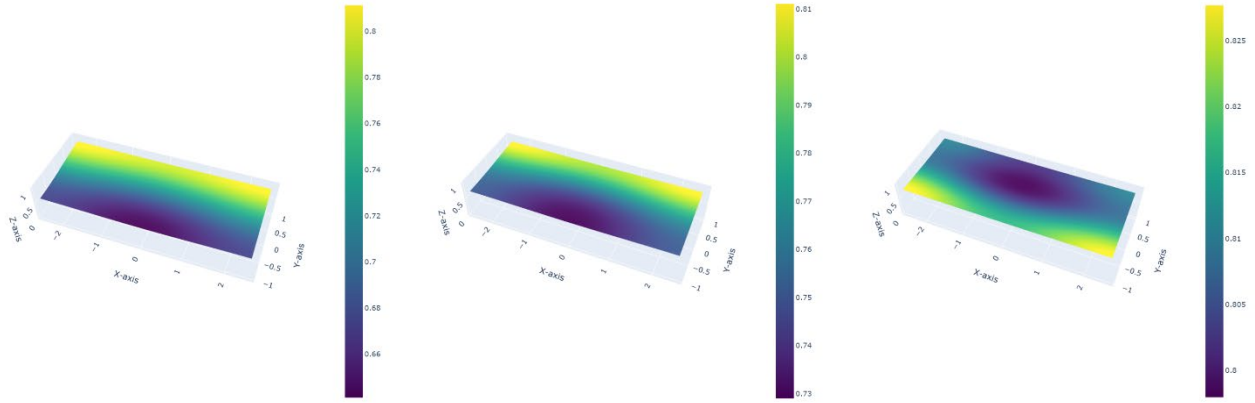


Figure 4: Simulated wear results (w=5.0dm, d=2.0dm, h=1.0dm, fn=20000/yr, year=200yr, material: granite, sole_material: rubber)

4.2 Travel pattern discrimination model

Here we will use two main methods: Random Forest Classifier and Finite Element Analysis (FEA).

Due to the difficulty of obtaining relevant training data, this process is omitted here, and only the workflow is provided.

4.2.1 Data Preprocessing for the Training Dataset:

The training data consists of 3D scanned images of stairs, represented as a two-dimensional function. However, since these functions generally lack continuity and other desirable properties, a finite element analysis (FEA) method is adopted to discretize the functional characteristics.

Divide the stair width into 100 segments and the stair depth into 10 segments, resulting in a total of 1000 rectangular finite elements $((w/100)*(d/10))$.

For each finite element, construct an array $height[1000]$. Perform integration of the function f within each finite element's range to calculate the volume V of the corresponding stair segment.

Compute the average height h_i of each finite element by dividing the volume V by the area of the finite element. Store h_i in $height[1000]$, where the index is uniquely determined by the finite element's position.

● **Advantages:** Dividing the stair surface into 1000 uniformly distributed finite elements ensures a more balanced consideration of the overall stair conditions. This approach effectively captures the height distribution across the entire stair surface. Furthermore, since walking patterns (e.g., bidirectional or unidirectional flow) are closely related to the wear levels of specific stair regions (i.e., height differences), the array $height[1000]$ can be used as the feature set for training the classifier.

4.2.2 Give the Walking Pattern Labels:

Table 2: Walking Pattern Labels

Depth\Width		1.25 δ (Single)		>2.5 δ (Pair)	
		D	L	D	L
>1.5 λ (Normal)	Upstairs	$\bar{X} \sim \mathcal{N}(\pm\delta/2, \sigma_1^2),$		$\bar{X} \sim \mathcal{N}(a/4 \pm \delta/2, \sigma_1^2),$	$\bar{X} \sim \mathcal{N}(a/4 \pm \delta/2, \sigma_1^2),$
		$\bar{Y} \sim \mathcal{N}(b/4, \sigma_2^2)$		$\bar{Y} \sim \mathcal{N}(b/4, \sigma_2^2)$	$\bar{Y} \sim \mathcal{N}(-b/2, \sigma_2^2)$
	Downstairs	$\bar{X} \sim \mathcal{N}(\pm\delta/2, \sigma_1^2),$	$\bar{X} \sim \mathcal{N}(\pm\delta/2, \sigma_1^2),$	$\bar{X} \sim \mathcal{N}(-a/4 \pm \delta/2, \sigma_1^2),$	$\bar{X} \sim \mathcal{N}(-a/4 \pm \delta/2, \sigma_1^2),$
		$\bar{Y} \sim \mathcal{N}(-b/4, \sigma_2^2)$	$\bar{Y} \sim \mathcal{N}(-b/2, \sigma_2^2)$	$\bar{Y} \sim \mathcal{N}(-b/4, \sigma_2^2)$	$\bar{Y} \sim \mathcal{N}(-b/2, \sigma_2^2)$
0.5 λ (Shallow)	Upstairs	$\bar{X} \sim \mathcal{N}(\pm\delta/2, \sigma_1^2),$		$\bar{X} \sim \mathcal{N}(a/4 \pm \delta/2, \sigma_1^2),$	$\bar{X} \sim \mathcal{N}(a/4 \pm \delta/2, \sigma_1^2),$
	Downstairs	$\bar{Y} \sim \mathcal{N}(b/4, \sigma_2^2)$		$\bar{Y} \sim \mathcal{N}(b/4, \sigma_2^2)$	$\bar{Y} \sim \mathcal{N}(-b/2, \sigma_2^2)$
		-		-	

$$\bar{X} \sim \mathcal{N}(-a/4 \pm \delta/2, \sigma_1^2),$$

$$\bar{Y} \sim \mathcal{N}(-b/2, \sigma_2^2)$$

4.2.3 Cross-validation and model application:

The dataset is divided into 10 subsets. Each time, 9 subsets are used as the training set, and the remaining 1 subset is used as the validation set. This process is repeated 10 times, with each subset serving as the validation set once. The final classification model is obtained by aggregating the results of these iterations. (Ehrlinger, 2015)

The trained model is then used to classify new stair data, assigning the corresponding walking pattern label based on the predicted category.

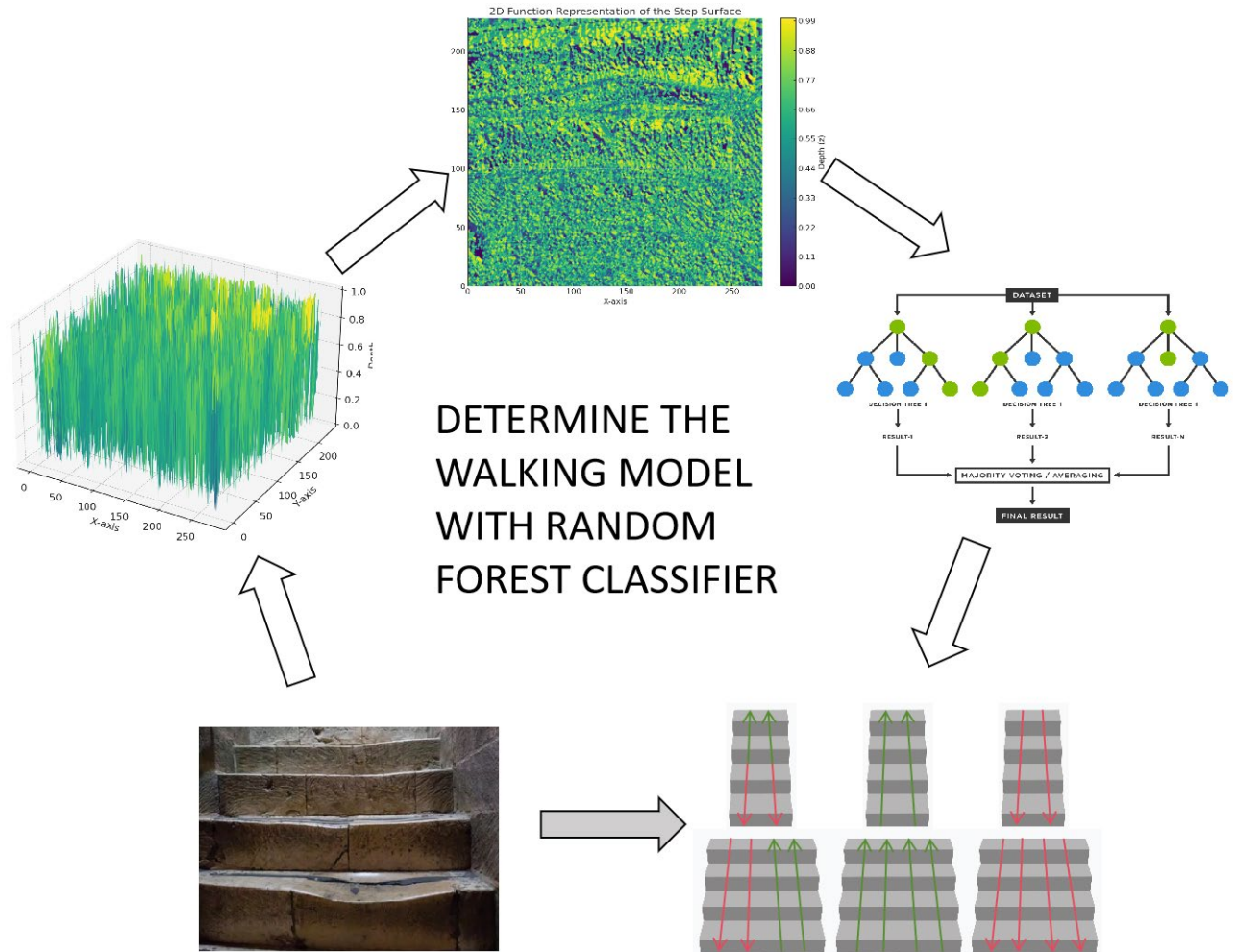


Figure 5: The flow picture of determining the walking model

4.3 Parameter optimizing model

We can use this model to predict some unknown parameters or make the obtained parameters more precise.

We Assume that the surface of the 3D-scanned stairs can be represented as a function $z = f_j(x, y)$. To estimate the formation parameters of f_j , a set of candidate fitting functions $g_i(x, y), i = 1, 2, 3, \dots$ is proposed. Residual analysis is employed to identify the best-fitting function g_i , which approximates the formation conditions of $z = f_j(x, y)$.

4.3.1 Model construction:

1) Random Sampling:

Randomly sample m points as $S = \{(x_k, y_k)\}_{k=1}^m$ from the domain of f_j , corresponding to the area of a single step.

2) Residual Calculation:

For each candidate function g_i , compute the sum of squared residuals (SSR) at the sampled points:

$$SSR(g_i) = \sum_{k=1}^m [f_j(x_k, y_k) - g_i(x_k, y_k)]^2 \quad (20)$$

3) Optimal Function Selection:

Compare the residuals for all candidate functions and select the function g_i that minimizes $SSR(g_i)$. Record the parameters of g_i , which may include factors such as pedestrian frequency, time period, and material properties.

4) Repeated Sampling and Averaging:

Repeat the random sampling process multiple times, storing the parameters obtained from each iteration. Calculate the average of these parameters to produce the final estimates. (Zhou, 2016)

4.3.2 Model application:

This approach corrects for potential biases introduced by individual sampling instances and ensures robust parameter estimation.

4.4 Age estimation and renovation determination model:

To estimate the construction age of stairs and determine whether they have undergone local renovations, the following method is proposed:

4.4.1 Some necessary assumptions:

● Assumption:

The age estimator (measured in years from the present) derived from residual minimization follows a normal distribution.

● Justification:

- Each stair has its age estimated by repeated simulations of the actual conditions. The estimate with the smallest residual is selected, and the average across all simulations is taken as the final estimator.

- Since the stair remains unchanged in each simulation, the residual-minimizing estimates are independent and identically distributed(i.i.d.).

- By the Central Limit Theorem, the average of a large number of i.i.d. estimates also follows a normal distribution.
- Hence, the final estimator follows $N(\mu, \sigma^2)$, where μ is the true construction age of the stair, and σ^2 is the variance of the estimator across stairs.
- **Key Hypothesis:**
 - Null hypothesis H_0 : The age estimates of all stairs follow the same normal distribution $N(\mu, \sigma^2)$, implying no local renovations.
 - Alternative hypothesis H_1 : The age estimates do not follow the same distribution, indicating local renovations.

4.4.2 Using Goodness-of-Fit Test to estimate construction age:

1) Moment Estimation:

Estimate μ as the sample mean of the age estimates.

Estimate σ^2 as the unbiased sample variance.

2) Interval Division:

To ensure that each interval contains at least 5 theoretical frequencies, divide the range of age estimates into k non-overlapping intervals $[x_1, x_2), [x_2, x_3), [x_3, x_4), \dots, [x_k, x_{k+1}]$.

3) Calculate Pearson's Statistic:

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

Where:

- O_i : Observed frequency in the i -th interval.
- E_i : Theoretical frequency in the i -th interval, computed using the assumed normal distribution: $E_i = n \cdot [F(x_{i+1}) - F(x_i)]$, with $F(x)$ as the cumulative distribution function of the assumed normal distribution and n as the total sample size. (DeGroot & Schervish, 2011)

4) Degrees of Freedom:

The degrees of freedom are $k - p - 1$, where $p = 2$ accounts for the two estimated parameters (mean and variance). So the degrees are $k - 3$.

5) Hypothesis Testing:

- Set the confidence level $\alpha = 0.95$.
- Use the χ^2 -distribution table, if $\chi^2 > \chi^2_{1-\alpha}(k-3)$, reject H_0 and conclude that the stairs have undergone local renovations. Otherwise, accept H_0 , indicating consistent construction age.

4.4.3 Use Post-analysis to determine renovation:

1) If no renovation:

Take μ , the sample mean of all age estimates, as the final construction age of the stairs.

2) If renovation is detected:

- Assume that stairs with more recent age estimates are likely renovated.
- Remove the smallest estimate from the dataset and repeat the analysis starting from Step 1 in section 4.4.2.

- If only one estimate remains, it is taken as the construction age of the earliest stairs.
- And we can know if there is whole renovation or partial renovation or both during this process.

When the error between each step is large enough, we can say partial renovation exists. And when the final μ has large enough difference with the practical construction age, we can say whole renovation exists. Undoubtedly, partial and whole renovation may exist at the same time.

3) Advantages

- Combines Pearson's statistic with moment estimation to assess the consistency of age estimates across stairs.
- Effectively determines whether local renovations have occurred.
- Provides robust estimates of the overall construction age when no renovations are present.
- Identifies the age of the earliest constructed stairs in cases of partial renovation.

5 Model Evaluation and Further Discussion

5.1 Sensitivity Analysis

In this article, we have developed several models. And in this part, we focus on the most significant model in our research: the holistic wear model.

This model relies heavily on parameters that may come from field surveys, online official databases or justified assumptions, such as the hardness of the steps material, the wear coefficient of the material and so on. Among them, the time interval, the passed time from built year to now, is significant because it has direct and indirect impact on both anthropogenic and natural wear remarkably. Thus, in this part, we keep other parameters fixed and vary time interval(years) from 50 to 300 in step size of 50. From the cross-sectional wear diagram, we can see that:

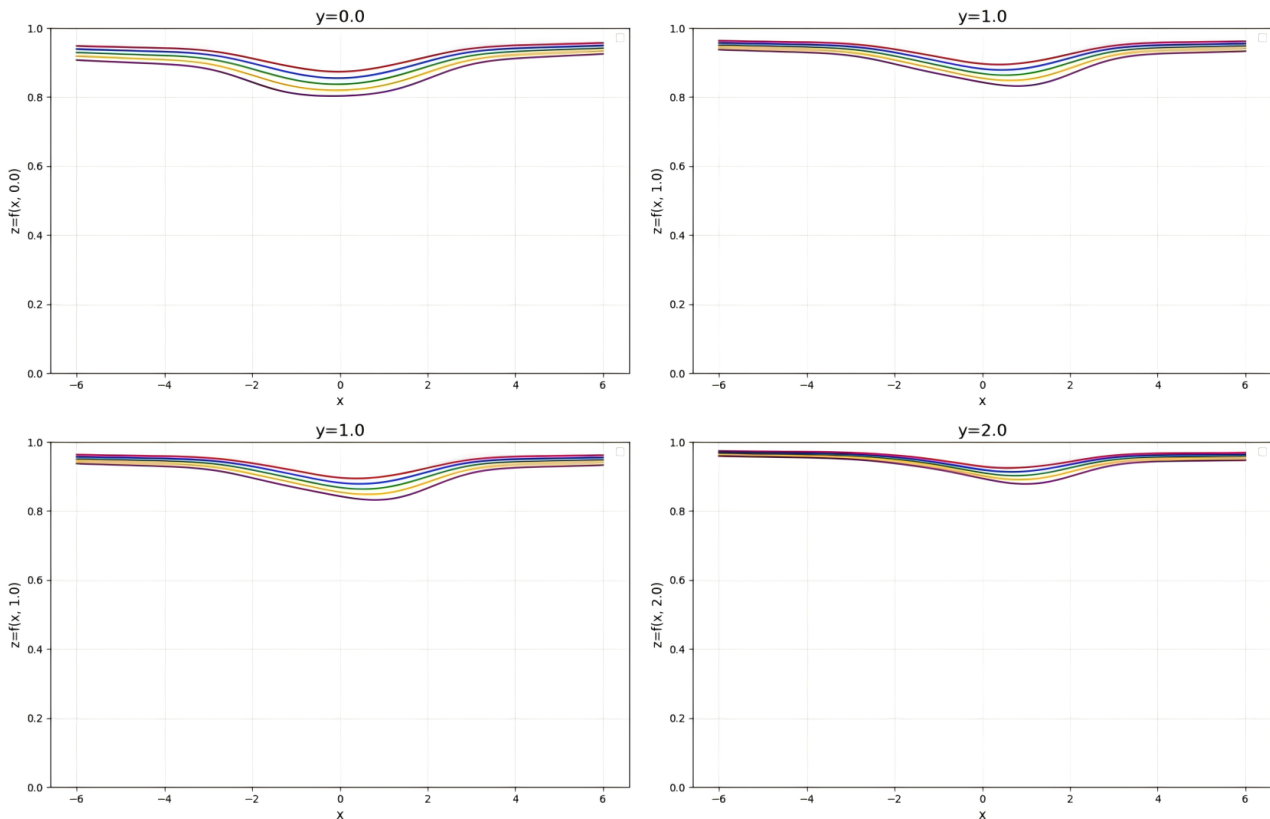


Figure 6: Cross-sectional wear diagram for sensitive analysis

The wear depth varies at all places although the difference may not be so remarkable. But this is just reasonable since wear is a longtime process. We only need to find that the wear depth change can be detected and the variation trend is reasonable. Thus, we can say the model corresponds to the time interval variation well and its reliability is good since no abnormal pattern appears. And our goal is to choose the most suitable predicted constructed time. And even small variation of time interval can be quantified in wear depth, which will make our result more precise and reliable.

5.2 Strengths

- Our method is non-destructive. Traditional methods like OSL method may need to destroy the steps and examine the materials below. However, our method only uses some non-destructive scanning methods and needs very little material to examine its composition to determine necessary parameters like wear coefficient.
- Our method is efficient. The steps we need are simplified and can be achieved by small teams. We only need to scan it, put the parameters into the model and then get the results.
- Our method is realistic. We consider not only anthropogenic factors, but also natural factors as well as coupling factors which make our holistic wear model more realistic.
- Our method is universal and expandable. Our model applies to different travel patterns, different step materials, different step shapes and so on. Also, our theory and model can be expanded to other relics to research their nature.

5.3 Weaknesses

- Our research only considers flat surface at initial state. However, there are steps with naturally uneven surfaces and steps with patterned surfaces. Distinguishing whether the unevenness results from the construction shape or wear is an unresolved issue. Additionally, for patterned steps, it is worth exploring whether the wear on the pattern can help us obtain necessary parameters or simplify the calculation process.

5.4 Further Discussion

For steps with patterns, such as those in the Forbidden City, we can focus on natural wear of the decorative patterned part in the center, as they are primarily symbolic and thus rarely walked upon. This provides a reference for studying the wear on the flat steps on either side and can function as parallel experiments to enhance the accuracy of the estimation. Also, as to this kind of steps, we can simplify the travel mode discrimination model since it usually determines the ascending and descending way naturally.

Also, we can expand our research to floors with patterns like those in Siena Cathedral in Italy. We can analyze the wear on these patterns to help determine the building's age. Also, we can try to recover the rallies that may have happened here in the past, which may help us understand the cultural background of the relics better.



Figure 7: The stairs in Forbidden City in Beijing, China(Left)



Figure 8: The floor in Siena Cathedral in Italy(Right)

6 Conclusion

In all, in this study, we used 3D scanning technology to survey the relevant shape data on the spot, discretized it through finite element analysis, determined the relevant walking mode through the random forest model, and optimized the remaining required data such as the properties of the steps after feasible analysis, and fitted the unknown data through the prediction model. After the data is input into the overall wear model, it is simulated, the goodness of fit test is carried out and continuously optimized. When a good fit is obtained, the year input at this time is used as the simulated year result. At the same time, it can be compared and analyzed with the year data obtained by other methods to determine the reliability of the results. At the same time, we can use the renovation detection model to perform self-analysis and overall analysis on the actual wear data to determine whether partial renovation and overall renovation have occurred.

This model will greatly facilitate archaeologists to simplify their work processes and improve work efficiency. It also has great scalability and is also of reference significance for studying cultural relics in other situations.

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